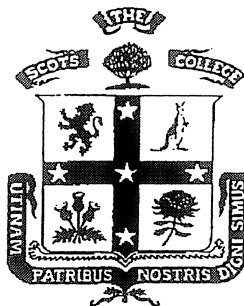


# The Scots College



TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION

2000

3/4 UNIT MATHEMATICS

Time Allowed :

TWO HOURS

*(Including 5 minutes reading time)*

Instructions to Candidates:

- ♦ All questions are to be attempted
- ♦ All questions are of equal value
- ♦ All necessary working should be shown for each question
- ♦ Non-programmable calculators that are Board approved are permitted
- ♦ A table of standard integrals is provided.

Booklet Order:

- Booklet 1: Questions 1 and 2  
Booklet 2: Questions 3 and 4  
Booklet 3: Questions 5, 6 and 7

### Question 1 (Begin a new booklet)

- (a) Evaluate  $\int_1^2 \frac{dx}{\sqrt{4-x^2}}$  (3 marks)
- (b) For  $y = -3 \sin^{-1} \frac{x}{2}$   
i) State the domain and range.  
ii) Sketch the curve. (3 marks)
- (c) If  $\tan \frac{\theta}{2} = t$ , express  $1 - \frac{1}{2} \sin \theta \tan \frac{\theta}{2}$  in terms of  $t$ . (3 marks)
- (d) Find the constants  $a, b$  such that  $x^2 - 2x - 3$  is a factor of the polynomial  
 $f(x) = x^3 - 3x^2 + ax + b$ . (3 marks)

### Question 2

- (a) Find  $\int \frac{x dx}{1+2x}$  using the substitution  $u = 1 + 2x$ . (3 marks)
- (b) (i) Show that the equation  $\log_e x - \cos x = 0$  has a root between  $x = 1$  and  $x = 2$ .  
(ii) By taking  $x = 1.2$  as the first approximation, use 1 step of Newton's method to find a better approximation to this root, correct to 2 decimal places. (3 marks)
- (c) Express  $3 \cos x + 4 \sin x$  in the form  $A \cos(x - \alpha)$  where  $A > 0$ . Hence, or otherwise, solve  $3 \cos x + 4 \sin x = -3$  for  $0 \leq x \leq 360^\circ$ . (4 marks)
- (d) Evaluate  $\int_0^{\frac{\pi}{2}} \cos^2 2x dx$ . (2 marks)

**Question 3 (Begin a new booklet)**

(a) Solve the equation  $2\ln(3x+1) - \ln(x+1) = \ln(7x+4)$

(3 marks)

(b)

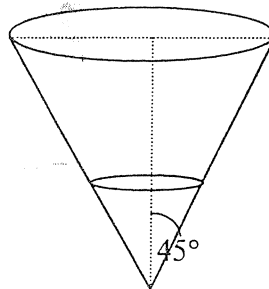


Figure not to scale

Water is being let into the conical vessel shown at a constant rate of  $8 \text{ cm}^3/\text{s}$ .

When the depth is 12 cm, find:

- (i) the rate of increase in the depth (in terms of  $\pi$ ), and
- (ii) the rate of increase in the area of the top surface of the water.

(5 marks)

(c) If  $\frac{P \sin A}{\tan B} = P \cos A + Q$ , show that  $P = \frac{Q \sin B}{\sin(A-B)}$ .

(3 marks)

(d) Differentiate with respect to  $x$ :  $y = \cos^{-1}(5x-4)$

(1 mark)

### Question 4

- (a) The acceleration of a particle moving in a straight line is given by :

$$\ddot{x} = 3 - 4x$$

where  $x$  is the displacement in metres, from the origin and  $t$  is the time in seconds.

If the particle starts from rest at  $x = 1$  metres,

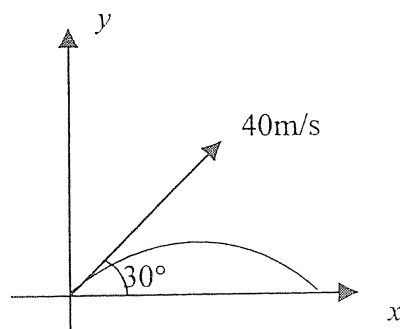
- (i) Show that the velocity of the particle is given by:

$$v^2 = 2(-2x^2 + 3x - 1)$$

- (ii) Identify the second position where the particle will come to rest.  
(iii) What will be the acceleration at the second position where the particle comes to rest?

(6 marks)

- (b) The diagram shows the path of an object launched at an angle of  $30^\circ$  to the horizontal with an initial speed of  $40\text{m/s}$  from O. The acceleration due to gravity is taken as  $10\text{m/s}^2$ , and air resistance is ignored.



- (i) Given that  $\frac{d^2x}{dt^2} = 0$  and  $\frac{d^2y}{dt^2} = -10$ , derive expressions for the horizontal displacement  $x(t)$  and the vertical displacement  $y(t)$  of the object from O,  $t$  seconds after launching.  
(ii) Using the equations in (i) above, calculate the time it takes for the object to land at A and the distance OA.

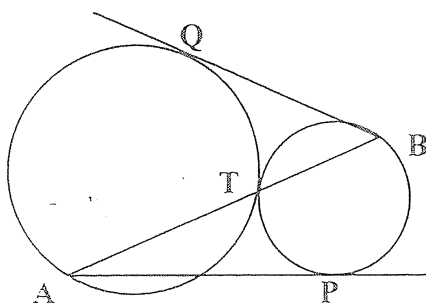
(6 marks)

### Question 5 (Begin a new booklet)

- (a) Use the method of Mathematical Induction to show that:  
 $n^3 + 2n$  is divisible by 3 for all positive integers  $n \geq 1$ .  
 (5 marks)
- (b) (i) Determine the equation of the tangent to the curve  $C: y = 2x^2$  at the point  $P(t, 2t^2)$ .
- (ii) The point  $Q$  lies on the curve  $C_1: y = x^2 + 1$ , on the same vertical line (ie with the same  $x$  coordinate) as the point  $P$  of part (i). Show that the equation of the tangent to  $C_1$  at  $Q$  is  $y = 2tx + (1 - t^2)$ .
- (iii) Find the precise locus of the points of intersection of these two tangents, as the common  $x$  coordinate  $t$  of the points  $P$  and  $Q$  assume all positive values. Indicate this locus on a sketch.  
 (7 marks)

### Question 6

- (a) Let  $f(x) = x^2 - 4x$  for all real  $x$ .
- (i) Explain why  $f(x)$  for all  $x \geq 2$  has an inverse function,  $f^{-1}(x)$ .
- (ii) State the domain and range of  $f^{-1}(x)$ .
- (iii) Find the coordinates of the point where  $y = f(x)$  and  $y = f^{-1}(x)$  meet.
- (iv) If  $0 < a < 2$  then find the value, in terms of  $a$ , of  $f^{-1}(f(a))$ .  
 (6 marks)
- (b)



The circles touch at  $T$ .  $ATB$  is a straight line.  $AP$  is a tangent to circle  $PTB$  and  $BQ$  is a tangent to circle  $QTA$ .

Prove that  $AP^2 + BQ^2 = AB^2$   
 (3 marks)

- (c) Show that  $\tan^{-1}\left(\frac{1}{4}\right) + \tan^{-1}\left(\frac{3}{5}\right) = \frac{\pi}{4}$

(3 marks)

### Question 7

(a) If  $\alpha, \beta$  and  $\gamma$  are the roots of the equation  $x^3 - 4x + 1 = 0$  evaluate:

(i)  $\alpha + \beta + \gamma$

(ii)  $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$

(3 marks)

(b) A beaker contains a coloured solution in which the amount of colouring,  $Q$ , is known to change at a rate given by  $\frac{dQ}{dt} = -0.02(Q - 30)$ . Initially the beaker contains 70g of colouring and  $t$  is in minutes.

(i) Write down an equation for  $Q$  in terms of  $t$ .

(ii) Find the amount of colouring, to the nearest gram, in the beaker after 45 minutes.

(3 marks)

(c) Assume that the tides rise and fall in Simple Harmonic Motion. A ship needs 10 metres of water to pass down a channel safely. At low tide the channel is 9 metres deep and at high tide the channel is 12 metres deep. Low tide is at 9 am and high tide is at 4 pm.

(i) State the period and amplitude of the motion.

(ii) Between what times can the ship be assured of safe passage?

(6 marks)

Question 1

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Solutions.

$$\int_1^2 \frac{dx}{\sqrt{4-x^2}} = \left[ \sin^{-1} \frac{x}{2} \right]_1^2$$

$$= \sin^{-1}(1) - \sin^{-1}\left(\frac{1}{2}\right)$$

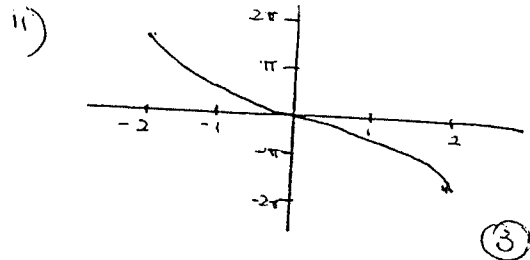
$$= \frac{\pi}{2} - \frac{\pi}{6}$$

$$= \frac{\pi}{3} \quad (3)$$

$$y = -3 \sin^{-1} \frac{x}{2}$$

i) Domain:  $-1 \leq \frac{x}{2} \leq 1$   
 $-2 \leq x \leq 2$

Range:  $+3\left(-\frac{\pi}{2}\right) \leq y \leq 3\left(\frac{\pi}{2}\right)$   
 $-\frac{3\pi}{2} \leq y \leq \frac{3\pi}{2}$



$$1 - \frac{1}{2} \left( \frac{2t}{1+t^2} \right) (t)$$

$$= 1 - \frac{2t^2}{2(1+t^2)}$$

$$= \frac{2+2t^2-2t^2}{2(1+t^2)}$$

$$= \frac{1}{(1+t^2)} \quad (3)$$

$$x^2 - 2x - 3 = (x-3)(x+1)$$

$(x-3)$  is a factor i.e.  $f(3) = 0$

$(x+1)$  is a factor i.e.  $f(-1) = 0$

$$f(x) = x^3 - 3x^2 + ax + b$$

$$f(3) = 27 - 27 + 3a + b = 0$$

$$3a + b = 0 \Rightarrow b = -3a$$

$$f(-1) = -1 - 3 - a + b$$

$$= -4 - a + b = 0$$

Sub  $b = -3a$

$$-4 - a - 3a = 0$$

Solution  $a = -1$

$$b = 3$$

(3)

## Question 2

$$\begin{aligned} 1) \int \frac{x}{1+2x} dx & \quad \text{let } u = 1+2x \Rightarrow x = \frac{u-1}{2} \\ & \quad du = 2dx \\ & \quad dx = \frac{1}{2} du \\ & = \int \frac{\frac{u-1}{2}}{u} \cdot \frac{1}{2} du \\ & = \int \frac{u-1}{4u} du \\ & = \int \frac{u}{4u} du - \int \frac{1}{4u} du \\ & = \int \frac{1}{4} du - \int \frac{1}{4u} du \\ & = \frac{1}{4} u - \frac{1}{4} \ln u + c \quad \text{but } u = (1+2x) \\ & = \frac{1+2x}{4} - \frac{\ln(1+2x)}{4} + c \quad (3) \end{aligned}$$

$$1) \log_e x - \cos x = 0$$

$$x=1 \quad \log_e 1 - \cos 1 = -0.54$$

$$x=2 \quad \log_e 2 - \cos 2 = 1.11$$

$\therefore$  a root lies between  $x=1$  (-ve answer) &  $x=2$  (ve answer)

$$11) z_1 = 1.2 \quad f(x) = \log_e x - \cos x$$

$$f'(x) = \frac{1}{x} + \sin x$$

$$z_2 = z_1 - \frac{f(z_1)}{f'(z_1)}$$

$$= 1.2 - \frac{f(1.2)}{f'(1.2)}$$

$$= 1.2 - \left( \frac{\log_e 1.2 - \cos 1.2}{\frac{1}{1.2} + \sin 1.2} \right) \quad (3)$$

$$= 1.30 \text{ (to 2 dec pl)}$$

$$3 \cos x + 4 \sin x$$

$$\sqrt{9+16} \cos(x-\alpha)$$

$$5 \cos(x-\alpha) \quad \tan \alpha = \frac{4}{3}$$

$$5 \cos(x-53^\circ 8') \quad \alpha = 53^\circ 8'$$

$$3 \cos x + 4 \sin x = -3$$

$$5 \cos(x-53^\circ 8') = -3$$

$$\cos(x-53^\circ 8') = \frac{-3}{5}$$

$$233^\circ 08'$$

$$x-53^\circ 8' = 126^\circ 57' \text{ or } (4)$$

Q. 2.2.2.2

$$\begin{aligned}
 &= \frac{1}{5} \int_0^{\frac{\pi}{2}} (1 + \cos 4x) dx \\
 &= \frac{1}{5} \left[ x + \frac{1}{4} \sin 4x \right]_0^{\frac{\pi}{2}} \\
 &= \frac{1}{5} \left[ \left( \frac{\pi}{2} + \frac{1}{4} \sin 2\pi \right) - \left( 0 + \frac{1}{4} \sin 0 \right) \right] \\
 &= \frac{\pi}{4}
 \end{aligned}$$

(2)

Question 3

$$2 \ln(3x+1) - \ln(x+1) = \ln(7x+4)$$

$$\ln(3x+1)^2 - \ln(x+1) = \ln(7x+4)$$

$$\ln \left( \frac{(3x+1)^2}{(x+1)} \right) = \ln(7x+4)$$

$$\frac{(3x+1)^2}{(x+1)} = (7x+4)$$

$$(3x+1)^2 = (7x+4)(x+1)$$

$$9x^2 + 6x + 1 = 7x^2 + 4x + 7x + 4$$

$$2x^2 - 5x - 3 = 0$$

$$(2x+1)(x-3) = 0$$

$$2x+1=0$$

$$\text{or } x-3=0$$

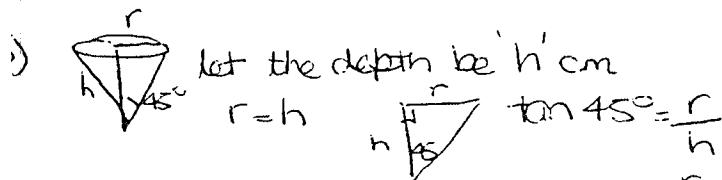
$$2x = -1$$

$$x = 3$$

$$x = -\frac{1}{2}$$

But  $\ln(3x+1)$  is not defined for  $x = -\frac{1}{2}$

$\therefore$  Solution  $x = 3$  (3)



$$1 = \frac{r}{h}$$

$$r = h$$

$$V = \frac{1}{3} \pi r^2 h$$

$$V = \frac{1}{3} \pi h^3$$

$$\frac{dV}{dh} = \pi h^2$$

$$\frac{dV}{dt} = \frac{dV}{dh} \times \frac{dh}{dt}$$

$$8 = \pi h^2 \times \frac{dh}{dt}$$

$$\therefore \frac{dh}{dt} = \frac{8}{\pi h^2}$$

$$= \frac{8}{\pi \times 12^2}$$

$$A = \pi r^2$$

$$\frac{dA}{dr} = 2\pi r$$

$$\frac{dA}{dt} = \frac{dA}{dr} \times \frac{dr}{dt}$$

$$= \frac{dA}{dh} \times \frac{dh}{dt}$$

$$= 2\pi r \times \frac{1}{12\pi}$$

$$= 24\pi \times \frac{1}{12\pi}$$

$$= \frac{1}{2} \text{ cm}^2/\text{s}$$

but as  $r = h$   
 $\frac{dr}{dt} = \frac{dh}{dt}$

$$1) \frac{P \cos A}{\tan B} = P \cos A + Q$$

$$P \cos A \times \frac{\cos B}{\sin B} = P \cos A + Q$$

$$P \cos A \cos B = P \cos A \sin B + Q \sin B$$

$$P \cos A \cos B - P \cos A \sin B = Q \sin B$$

$$P(\cos A \cos B - \cos A \sin B) = Q \sin B$$

$$P(\cos(A-B)) = Q \sin B$$

$$P = \frac{Q \sin B}{\cos(A-B)}$$

(3)

$$y = \cos^{-1}(5x-4)$$

$$\frac{dy}{dx} = \frac{1}{\sqrt{1-(5x-4)^2}} \times 5$$

$$= \frac{5}{\sqrt{1-(25x^2-40x+16)}}$$

$$= \frac{5}{\sqrt{40x-25x^2-15}}$$

(1)

Question 4

$$\ddot{x} = 3-4x$$

$$\frac{d}{dx} \left( \frac{1}{2} v^2 \right) = 3-4x$$

$$\frac{1}{2} v^2 = \int (3-4x) dx$$

$$= 3x - 2x^2 + C$$

$$\text{when } t=0, x=1, v=0$$

$$0 = 3 - 2 + C$$

$$C = -1$$

$$\frac{1}{2} v^2 = 3x - 2x^2 - 1$$

$$v^2 = 2(3x - 2x^2 - 1)$$

When the particle comes to rest  $v=0$

$$0 = -2(2x^2 - 3x + 1)$$

$$2x^2 - 3x + 1 = 0$$

$$(2x-1)(x-1) = 0$$

$$x = \frac{1}{2} \text{ or } x = 1$$

The particle will come to rest again

$$\text{when } x = \frac{1}{2} \text{ m.}$$

$$\ddot{x} = 3 - 4x \text{ when } x = \frac{1}{2}$$

$$\ddot{x} = 3 - 2$$

(6)

$$\frac{d^2x}{dt^2} = 0$$

$$\ddot{x} = 0$$

$$\dot{x} = C$$

$$\text{when } t=0, \dot{x} = V \cos \theta$$

$$\therefore C = V \cos \theta$$

$$\dot{x} = V \cos \theta$$

$$x = Vt \cos \theta + C$$

$$\text{when } t=0, x=0$$

$$\therefore C=0$$

$$x = Vt \cos \theta$$

$$x = (40 \cos 30^\circ)t$$

$$= 20\sqrt{3}t$$

vertical

$$\ddot{y} = -10$$

$$\dot{y} = -10t + C$$

$$\text{when } t=0, \dot{y} = V \sin \theta$$

$$\therefore C = V \sin \theta$$

$$\dot{y} = V \sin \theta - 10t$$

$$y = -5t^2 + Vt \sin \theta + C$$

$$\text{when } t=0, y=0$$

$$\therefore C=0$$

$$y = -5t^2 + 40 \sin 30^\circ t$$

$$= -5t^2 + 20t$$

$$1) y=0 \quad -5t^2 + 20t = 0$$

$$-5t(t-4) = 0$$

$$t=0 \text{ or } t=4 \text{ sec}$$

$\therefore$  object lands after 4 sec

$$\text{when } t=4, x = 20\sqrt{3} \times 4$$

$$= 80\sqrt{3} \text{ m}$$

$$\therefore OA = 80\sqrt{3} \text{ m}$$

(6)

Question 5

$$\text{Let } n^3 + 2n = 3P$$

$$1=1 \quad 1+2=3 \quad \therefore \text{true for } n=1$$

assume it true for  $n=k$

$$\text{i.e. } k^3 + 2k = 3P$$

now true for  $n=(k+1)$

$$\text{so } (k+1)^3 + 2(k+1)$$

$$= k^3 + 3k^2 + 3k + 1 + 2k + 2$$

$$= k^3 + 2k + 3k^2 + 3k + 3$$

$$= 3P + 3(k^2 + k + 1)$$

$$= 3[P + (k^2 + k + 1)]$$

$$= 3m$$

$$\text{for } m = [P + (k^2 + k + 1)]$$

$\therefore$  divisible by 3

is true for  $n=1$ , &  $n=k$  & proven for  $n=k+1$

then true for all  $n > 0$

(5)

$$1) y = 2x^2$$

$$\frac{dy}{dx} = 4x \text{ at } x=t$$

$$m = 4t$$

$$m = \frac{y - y_1}{x - x_1}$$

$$\frac{4t}{1} = \frac{y - 2t^2}{x - t}$$

$$y - 2t^2 = 4t(x - t)$$

$$y - 2t^2 = 4tx - 4t^2$$

$$y = 4tx - 2t^2$$

$$1) y = x^2 + 1$$

$$\frac{dy}{dx} = 2x \text{ at } x=t$$

$$m = 2t$$

$$y = t, y = (t^2 + 1)$$

$$m = \frac{y - y_1}{x - x_1}$$

$$\frac{2t}{1} = \frac{y - (t^2 + 1)}{x - t}$$

$$y - (t^2 + 1) = 2t(x - t)$$

$$y - (t^2 + 1) = 2tx - 2t^2$$

$$y = 2tx - t^2 + 1$$

$$y = 2tx + (1 - t^2)$$

$$1) y = 4tx - 2t^2$$

$$y = 2tx + (1 - t^2) \quad \left. \begin{array}{l} \text{Solve} \\ \text{Simultaneously} \end{array} \right\}$$

$$4tx - 2t^2 = 2tx + (1 - t^2)$$

$$2tx = 2t^2 - t^2 + 1$$

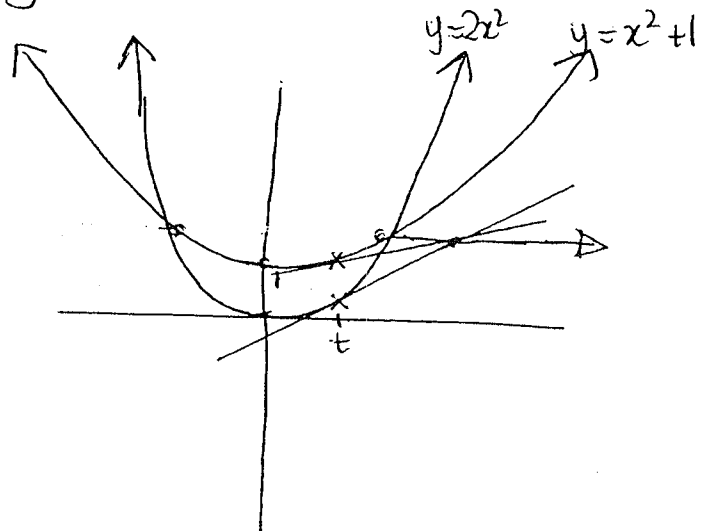
$$2tx = t^2 + 1$$

$$x = \frac{t^2 + 1}{2t}$$

$$y = 4t \left( \frac{t^2 + 1}{2t} \right) - 2t^2$$

$$y = 2(t^2 + 1) - 2t^2$$

$$y = 2$$



required locus is  $y=2$

where  $x > 1$  &  $x < -1$

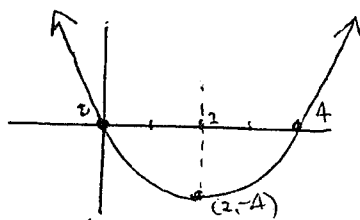
as  $x$  must be positive only  $x > 1$

(6)

# Question 6

$$1) f(x) = x^2 - 4x$$

$$= x(x-4)$$



i) for every y there is one & only one x.

ii) Domain  $x \geq -4$

Range  $y \geq -4$

iii)  $y = x^2 - 4x$   $x = y^2 - 4y$

$$y = (y^2 - 4y)^2 - 4(y^2 - 4y)$$

$$y = y^4 - 8y^3 + 16y^2 - 4y^2 + 16y$$

$$0 = y^4 - 8y^3 + 12y^2 + 15y$$

$$y(y^3 - 8y^2 + 12y + 15) = 0$$

$$p(x) = y^3 - 8y^2 + 12y + 15$$

$$p(3) = 27 - 72 + 36 + 15$$

$$= 6$$

$$p(5) = 125 - 200 + 60 + 15$$

$$= 0$$

$\therefore (y-5)$  is a factor

$$f^{-1}(f(a))$$

$$= 2 + \sqrt{f(a) + 4}$$

$$= 2 + \sqrt{a^2 - 4a + 4}$$

$$= 2 + \sqrt{(a-2)^2}$$

$$= 2 + (\sqrt{a-2})^2$$

$$= 2 + |a-2|$$

$AP^2 = AB \cdot AT$  (square of tangent equals product of secant segments)

$BQ^2 = AB \cdot BT$  ( " " " " )

$$P^2 + BQ^2 = AB \cdot AT + AB \cdot BT$$

$$= AB(AT + BT)$$

$$= AB \cdot AB$$

$$= AB^2$$

$$\begin{array}{r} y^2 - 3y - 3 \\ y-5 \overline{) y^3 - 8y^2 + 12y + 15} \\ \underline{y^3 - 5y^2} \phantom{+ 15} \\ -3y^2 + 12y \phantom{+ 15} \\ \underline{-3y^2 + 15y} \phantom{+ 15} \\ -3y + 15 \\ \underline{-3y + 15} \\ 0 \end{array}$$

$$y(y-5)(y^2-3y-3) = 0$$

$$y=0 \text{ or } y=5 \text{ or } y^2-3y-3=0$$

$$\therefore x = 25 - 20$$

$$x = 5$$

$$(5, 5)$$

⑥

③

$$\tan^{-1} \frac{1}{4} = \alpha$$

$$\tan \alpha = \frac{1}{4} \quad -\frac{\pi}{2} < \alpha < \frac{\pi}{2}$$

$$\text{Let } \tan^{-1} \frac{3}{5} = \beta$$

$$\tan \beta = \frac{3}{5} \quad -\frac{\pi}{2} < \beta < \frac{\pi}{2}$$

$$\tan \left[ \tan^{-1} \frac{1}{4} + \tan^{-1} \frac{3}{5} \right]$$

$$= \tan(\alpha + \beta)$$

$$= \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$$

$$= \frac{\frac{1}{4} + \frac{3}{5}}{1 - \frac{1}{4} \times \frac{3}{5}}$$

$$= \frac{\frac{17}{20}}{\frac{17}{20}}$$

$$= 1$$

$$\tan \left[ \tan^{-1} \left( \frac{1}{4} \right) + \tan^{-1} \left( \frac{3}{5} \right) \right] = 1$$

$$\tan^{-1}(1) = \tan^{-1} \left( \frac{1}{4} \right) + \tan^{-1} \left( \frac{3}{5} \right)$$

$$\tan^{-1} \left( \frac{1}{4} \right) + \tan^{-1} \left( \frac{3}{5} \right) = \frac{\pi}{4}$$

(3)

Question 7

$$3-4x+1=0 \quad \begin{aligned} a &= 1 \\ b &= 0 \\ c &= -4 \\ d &= 1 \end{aligned}$$

$$\alpha + \beta + \gamma = -\frac{b}{a}$$

$$= 0$$

$$\begin{aligned} 4\beta + \alpha\gamma + \beta\gamma &= \frac{c}{a} \\ &= -\frac{4}{1} \\ &= -4 \end{aligned}$$

$$\begin{aligned} 4\beta\gamma &= -\frac{d}{a} \\ &= -\frac{1}{1} \\ &= -1 \end{aligned}$$

$$\begin{aligned} \text{ii) } \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} &= \frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma} \\ &= \frac{-4}{-1} \\ &= 4 \end{aligned}$$

(3)

$$\frac{dQ}{dt} = -0.02(Q - 30)$$

$$Q = 30 + Ae^{-0.02t}$$

$$t=0, Q=70$$

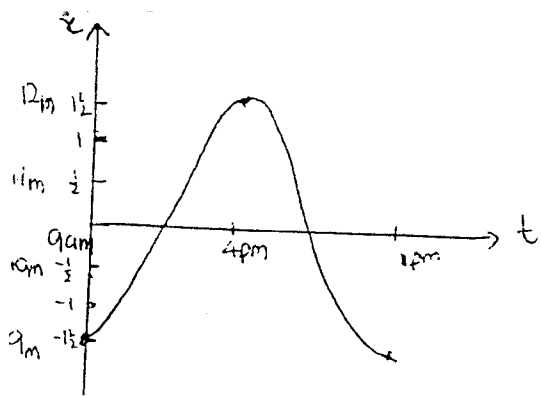
$$70 = 30 + A$$

$$Q = 30 + 40e^{-0.02t}$$

$$\text{ii) } t=45 \quad Q = 30 + 40e^{-0.02 \times 45}$$

$$\approx 46g$$

(3)



Let  $12m \Rightarrow 1\frac{1}{2}m$  or  $x$  axis  
 $9m \Rightarrow -1\frac{1}{2}m$  or  $x$  axis

$$i) \text{ Period} = \frac{2\pi}{\omega}$$

$$A = \frac{2\pi}{\omega}$$

$$\omega = \frac{\pi}{1}$$

$$\text{Amplitude} = 1\frac{1}{2}m$$

$$\therefore \text{Period} = 2 \text{ hrs}$$

$$\therefore x = a \cos(\omega t + \alpha)$$

$$x = -1\frac{1}{2} \cos\left(\frac{\pi}{1}t + \alpha\right)$$

$$\text{her } t=0, x=1\frac{1}{2}$$

$$-1\frac{1}{2} = -1\frac{1}{2} \cos \alpha$$

$$\cos \alpha = 1$$

$$\alpha = 0$$

$$\text{her } x = -\frac{1}{2}$$

$$-\frac{1}{2} = -1\frac{1}{2} \cos\left(\frac{\pi}{1}t\right)$$

$$\cos\left(\frac{\pi}{1}t\right) = \frac{1}{3}$$

$$\frac{\pi}{1}t = 1.2309 \dots$$

$$t = 2.74$$

$$= 2 \text{ hrs } 45 \text{ min}$$

$$\therefore 9 \text{ am} + 2 \text{ hrs } 45 \text{ min}$$

$$= 11.45 \text{ am}$$

$$11 - 2 \text{ hr } 45 \text{ min}$$

$$= 8.15 \text{ pm}$$

⑥